

# TRIAGE-LINK

## Offline-first PWA for field casualty documentation & coordination

Capability overview — the PWA, the encryption & audit layer, the hosted multi-tenant backend with admin security, and the market flavors (Humanitarian / NGO · Ontario EMS / regulated · productized backend).

React + TypeScript · IndexedDB + op-log · WebCrypto · Fastify · NEMESIS/OADS

**TRIAGE-LINK** offline-first field casualty documentation

# One field record, no signal required

- ▶ Document a casualty completely OFFLINE — injuries, vitals, treatments, photos, triage, handover — with no server in the loop.
- ▶ Built for where connectivity is unreliable or absent: disaster/MCI, humanitarian field clinics, mass-gathering and prehospital EMS.
- ▶ Installs as a PWA; data lives locally in IndexedDB and survives reloads, crashes, and power cycles.
- ▶ Optional, never-required sync aggregates records across a team only where a deployment wants it.
- ▶ One codebase, three market 'flavors' that share the offline-first core.

# Offline-first core, optional everything else

## DEVICE – PWA

- React + TS UI
- IndexedDB (Dexie)
- Op-log sync engine
- WebCrypto vault
- Operators + audit



## OPTIONAL – Sync service

- Fastify, multi-tenant
- Conflict-aware /sync
- OIDC admin + console
- Quota · retention · limits
- PostgreSQL



## OPTIONAL – EHR gateway

- ONE ID / Ontario Health
- PCR \$match · handover
- mTLS, server-side
- NEMESIS/OADS export
- Audit trail

The PWA is fully functional with neither optional tier present. Sync and the EHR gateway are additive, server-side, and off by default.

# Documenting a casualty

## Body & injuries

- Tap-to-mark injuries on an anterior/posterior body chart
- Injury palette (GSW, burn, laceration, fracture...) + severity
- Per-injury notes + wound photos (stored as out-of-line blobs)
- Burn TBSA auto-estimate (Lund-Browder by age band)

## Triage & identity

- START-style tag: Immediate / Delayed / Minor / Deceased
- Patient: name, DOB→age band, sex, MRN, NOK, blood type
- Incident: time, mechanism, location
- Color-coded multi-casualty Triage Board

# The field record — one screen, fully offline

The screenshot displays the TRIAGE-LINK field casualty record interface, designed for offline use on a mobile device. The interface is organized into several sections:

- Header:** Includes the application name "TRIAGE-LINK", the title "FIELD CASUALTY RECORD", and a case identifier "CAS-VHIAFGJR". It also features navigation links for "FR", "New casualty", "Board · 1", "Summary", "Tour", and "Language pack".
- Actions:** A row of buttons for "Language template", "Operators", "Audit log", "Send to EHR ↑", "Encrypt data", and "Export FHIR ↓".
- Triage Status:** A horizontal bar with tabs for "Immediate" (selected), "Delayed", "Minor", and "Deceased". A red indicator "Immediate (Red)" is visible on the right.
- TOMBSTONE — IDENTITY:** A section for patient identification with fields for "FULL NAME" (Surname, Given), "DATE OF BIRTH" (YYYY-MM-DD), "SEX" (dropdown), "PATIENT ID / MRN" (CAS-VHIAFGJR), "BLOOD TYPE" (Unknown), "NEXT OF KIN", and "NOK PHONE". A button "Verify identity against PCR" is at the bottom.
- INCIDENT:** A section for incident details with fields for "TIME OF INJURY" (mm/dd/yyyy, --:-- --), "MECHANISM" (Blunt, RTC, GSW...), "LOCATION OF INCIDENT" (Address / grid / GPS), and "AGE BAND" (dropdowns for <1y, 1y, 5y, 10y, 15y, Adult).
- INJURY CHART — ANTERIOR / POSTERIOR:** A section for injury documentation with buttons for "Fracture", "Laceration", "Burn" (selected), "Gunshot", "Contusion", "Amputation", "Abrasion", and "Puncture". It includes a "Full body" image and a "POSTERIOR · BACK" image.
- ACUITY AT A GLANCE:** A section for quick assessment with a red "Immediate (Red)" tag, "LATEST VITALS" (GCS 14 (E4 V4 M6), Jun 28, 02:17 PM), and "1 injury" with a "4.5% TBSA" tag.
- LOGGED INJURIES:** A section for injury history with a "4.5% TBSA" tag and a list of injuries, including "Burn · moderate" (R Chest · anterior). It includes buttons for "Minor", "Moderate" (selected), "Severe", and "Critical", and a "Notes" field.

Triage tag · patient identity · incident · injury body-chart · acuity glance (GCS, TBSA) · vitals — captured on-device.

# Vitals, interventions & trends

## Vitals & scoring

- Timestamped vital sets: HR, BP, RR, SpO<sub>2</sub>, GCS, pain
- Built-in GCS calculator (eye/verbal/motor → total)
- Vitals-trend sparklines once  $\geq 2$  readings exist
- Time-since-injury clock (T+) on the record

## Treatments

- Structured log: tourniquet, airway, decompression, IV/fluids,
- medication, splinting, wound packing, CPR...
- Each entry timestamped + attributed to the on-duty operator
- Feeds the AT-MIST handover summary

# Summaries & clean handoff

- ▶ One-page printable Casualty Summary card (AT-MIST) — print or save as PDF for handover.
- ▶ Scene Summary / command roll-up: casualties tallied by triage, on-scene vs handed-over.
- ▶ Handover sign-off (who took over care, facility, time) emitted as a FHIR handover bundle.
- ▶ Optional 'Send to EHR' contributes the handover to a provincial EHR via the gateway.
- ▶ Everything renders offline; nothing leaves the device unless you export or sync.

# Handover card & the scene picture

TRIAGE-LINK FIELD CASUALTY RECORD CASE CAS-VIWBXICJ [Print / Save PDF](#) [Close](#) Board · 1 [Summary](#) [? Tour](#) [+ Language pack](#)

Language template

Amputation Abrasion

AT-MIST HANDOVER

**A** Age / sex — Adult  
**T** Time of incident — —  
**M** Mechanism — —  
**I** Injuries — R Chest burn (moderate)  
**S** Signs — GCS 14 (E4 V4 M6)  
**T** Treatment — —

PATIENT

|             |              |            |       |
|-------------|--------------|------------|-------|
| Name        | —            | DOB        | —     |
| Sex         | —            | Age band   | Adult |
| MRN         | CAS-VIWBXICJ | Blood type | —     |
| Next of kin | —            | NOK phone  | —     |
| Address     | —            |            |       |

INCIDENT

|                |   |           |   |
|----------------|---|-----------|---|
| Time of injury | — | Mechanism | — |
| Location       | — |           |   |

INJURIES 1

| REGION  | VIEW     | TYPE | SEVERITY | NOTES |
|---------|----------|------|----------|-------|
| R Chest | anterior | Burn | moderate | —     |

Burn TBSA: 4.5% (Lund-Browder, Adult)

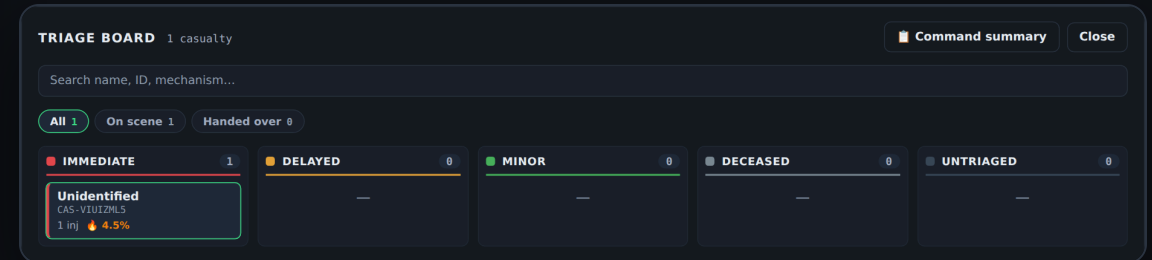
VITALS 1

| TIME                   | HR | BP | RR | SPO <sub>2</sub> | GCS           | PAIN |
|------------------------|----|----|----|------------------|---------------|------|
| Jun 28, 2026, 02:18 PM | —  | —  | —  | —                | 14 (E4 V4 M6) | —    |

TREATMENTS 0

None recorded.

One-page AT-MIST casualty card — print or save as PDF.



Triage Board — every casualty by acuity, on-scene vs handed-over.



# Four languages, guided & spoken

## Internationalization

- EN / FR / AR / FA built in — Arabic & Persian fully RTL
- Loadable JSON language packs: add a language with NO app release
- Downloadable English template to translate; parity-tested in CI
- Natural wording, not literal — reviewed per language

## Guided tour

- Smart tour highlights each real control
- Offline voice-over (SpeechSynthesis) in the active language
- Action steps auto-advance once the user does them
- Every user-visible feature is taught in the tour (enforced)

# Installable, offline, durable

- ▶ Installable PWA (service worker + Workbox precache) — launches and runs with no network.
- ▶ All data in IndexedDB via Dexie; records, op-log, photos, audit chain persist locally.
- ▶ Phones, tablets, laptops; responsive layout collapses gracefully on small screens.
- ▶ No telemetry, no implicit network calls — a casualty is documented entirely on-device.

# Conflict-aware op-log sync engine

- ▶ Every change is journaled as an immutable operation (scalars + collections), not a blind overwrite.
- ▶ Deterministic resolve(): Lamport clocks order edits; ties break predictably — same inputs, same result, every device.
- ▶ Concurrent edits to different fields all survive; same-field edits pick a deterministic winner and REPORT the conflict (losing op retained).
- ▶ The server stores and folds ops — it does not implement its own divergent merge logic.
- ▶ Incremental sync (cursor) pulls only what changed; full-state pulls are paginated.

# Opt-in encryption vault

## How it works

- AES-256-GCM, key derived from a passphrase via PBKDF2 (210k iters)
- Encrypts the heaviest PHI — wound photos — plus records & op-log
- Key lives only in memory while unlocked; locking drops it
- Idle auto-lock; wrong passphrase rejected via a verifier

## Safety posture

- DEFAULT-OFF: with no vault, behavior is byte-for-byte unchanged
- Mixed plaintext/encrypted rows read correctly through a toggle
- Sealed records unreadable (get/list skip them) while locked
- Crash-safe enable/disable — data is never orphaned

# Operators, RBAC-lite & tamper-evident audit

## Shared-device access

- Local operator roster (field / lead / admin)
- Records & audit entries attributed to the on-duty operator
- Step-up PIN re-auth gates sensitive actions (delete, export...)
- Empty roster = open (community default); adding operators opts in

## Audit log

- Append-only, hash-chained entries (SHA-256 per entry)
- Tampering/deletion breaks the chain — detectable offline
- No update/delete API; reviewable even while the vault is locked
- Covers create / view / delete / export / vault / step-up

# Backup, restore & export

## Backup / restore

- Full JSON backup of every record (plain or encrypted)
- Restore by merge (keep newer of duplicates) or replace
- Encrypted backup keeps PHI unreadable without the passphrase

## CSV interchange

- Roster CSV export (identity + incident) for analytics/QA
- CSV import onboards a patient list from paper/another system
- Date-range filter scopes an export to a window
- Deployment provenance stamped on rows (humanitarian)

# Multi-tenant sync service (Fastify)

- ▶ Optional cloud or self-hosted backend for cross-team aggregation — the PWA never requires it.
- ▶ Per-tenant isolation: each API key authenticates AND scopes a tenant's data; conflict-aware /sync stores and folds ops.
- ▶ Hardened: sanitized error envelopes, paginated pulls, per-tenant storage quota, audit-log retention TTL.
- ▶ OpenAPI 3 + Swagger UI; liveness/readiness probes; per-tenant metrics; request-id correlation.
- ▶ Runs on PostgreSQL; graceful shutdown; rate-limited per IP.

# Admin security & console

## Admin authentication

- Tenant-admin API (/admin/\*) behind a static token OR OIDC SSO
- OIDC: IdP-issued JWT, audience-checked, optional role mapping
- Provision tenants; issue / rotate / revoke per-tenant API keys
- Every admin mutation written to a separate admin-audit trail

## Graphical console

- Opt-in static admin console at /console
- Token-entry auth; holds no secrets (the API gate enforces)
- Browse tenants, keys, metrics & audit from a browser
- Off by default — enabled per deployment



THREE MARKET FLAVORS

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# One core, productized three ways

Humanitarian / NGO · Ontario EMS (regulated) · Productized backend

# Field documentation where the cloud isn't

## Shipped on this line

- Deployment context — device-wide operation tag + provenance banner
- Disaster/MCI mode — one toggle makes encryption mandatory + command roll-up
- Kiosk defaults — assign-operator prompt + 2-min auto-lock on shared devices
- Donor export — provenance-stamped CSV + date-range filter

## More

- Retention presets — confirmed, step-up-gated purge window (off/30/90/180/365 d)
- Air-gapped packaging — one Docker stack (PWA + sync + Postgres) offline on a laptop
- Loadable language packs for any region; encrypted backups for low-infra handoff
- Intended use = documentation/coordination tool (light reg load)

# Toward a conformant ePCR

## NEMSIS / OADS conformance

- Section exporter maps captured data → NEMSIS v3.5 / OADS v4.0 shapes
- Deterministic offline XML serializer (shaped, gap-annotated)
- Pluggable validator (cardinality/datatype/value-set) + placeholder ruleset
- Capture panels: eResponse / eTimes / eCrew / eScene — gaps clear at runtime

## In-app & integration

- Read-only Conformance view: live gaps + validator issues, in 4 languages
- Clearly labelled NOT certification (placeholder ruleset, rulesetSource shown)
- ONE ID / Ontario Health gateway scaffold (mTLS, server-side) for PCR \$match
- Gated next: official-dictionary reconciliation + live ONE ID credentials

# The shared, hardened service line

- ▶ The multi-tenant sync service + admin security is the common spine both market branches reuse.
- ▶ Multi-tenant isolation, OIDC SSO + role-based admin, per-tenant rate limits & quota, incremental sync.
- ▶ EHR-access and admin audit trails — necessary for any regulated, multi-service deployment.
- ▶ Cascaded to every flavor branch, so a hardening fix lands everywhere.

# Where things stand

## Done & merged

- Full offline PWA: capture, triage, vitals, handover, summaries
- Encryption vault · operators · step-up · tamper-evident audit
- Multi-tenant backend hardening + graphical admin console
- Humanitarian: all 5 roadmap items (context→MCI→export→retention→air-gap)
- Ontario: NEMSIS pipeline + capture panels + in-app conformance view

## Externally gated next

- Ontario PR-2c — official NEMSIS/OADS dictionary + true XSD validation
- Ontario PR-4 — live ONE ID credentials + real mTLS client cert
- SOC 2 Type II / QMS / SaMD evidence (if intended use crosses the device line)
- CAD / hospital-EHR handover partnerships

# Document anywhere.

Offline-first by design · encrypted & audited · ready for the cloud only when you want it.

TRIAGE-LINK — PWA · multi-tenant backend · Humanitarian & Ontario EMS flavors